

TIC 437150434

Eclipsing binary candidate dossier - Star Vetter

CLEAN

BLS EPHEMERIS

Period 5.285343 d
Epoch (t0) 2230.8422 BTJD
Duration 0.1700 d
Depth 15.003 % (150,033 ppm)
SNR 1081.0

INDEPENDENT CHECKS (f3i gates)

Eclipses observed 4 (gate ≥ 3)
Robust depth ratio 1.18 (gate ≥ 0.50)
Robust depth sigma 891.6 (gate ≥ 7.0)
Odd/even z-score 1.10 (consistent)
Secondary eclipse not detected

CATALOG CROSS-MATCH (novelty)

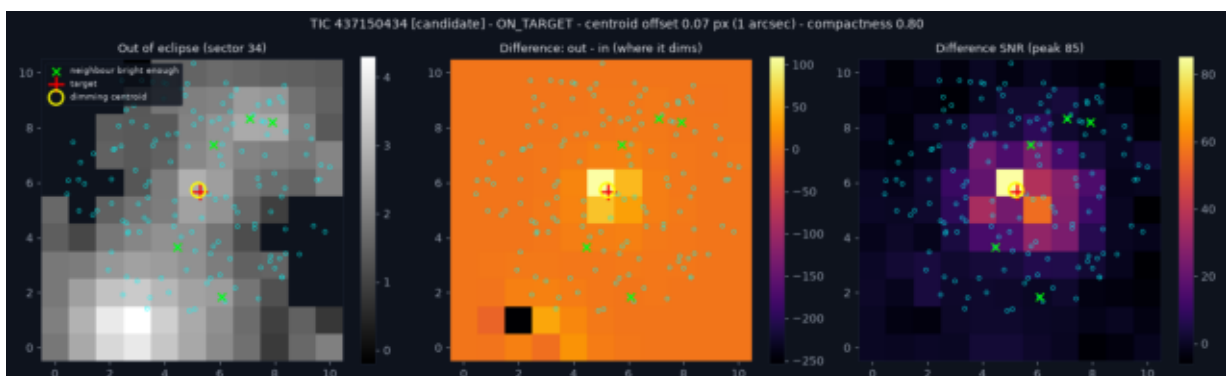
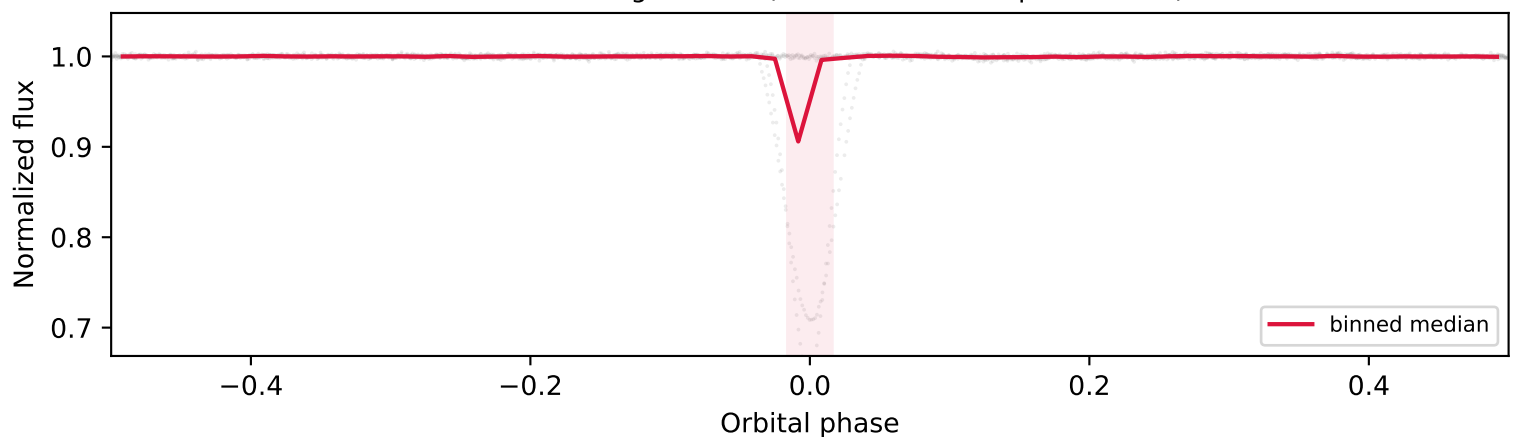
VSX no match
ASAS-SN no match
Gaia eclipsing-binary table no match
Gaia variable classification no match
TESS EB catalog (Prsa+2022) no match
ExoFOP TOI list no match

Verdict: NOVEL

PIXEL-LEVEL SOURCE CHECK (TESScut difference imaging)

Sector 34 light-curve dip in this sector: 645.4 sigma t0 refined by -0.0075 phase
Difference-image peak SNR 85.33 (gate ≥ 5.0) Compactness 0.80 (gate ≥ 0.50)
Dimming centroid offset 0.07 px (1 arcsec) (gate ≤ 1.0 px, i.e. on the target)
Nearest bright-enough neighbour in cutout: TIC 437150402
1.75 px away, dTmag +1.46 - farther from the dimming than the target

Phase-folded light curve (shaded = BLS eclipse window)



No caveats - clean detection on every check.